

## CO5 - AT<sub>1</sub> - Pseudocode

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### Social Network Analysis.

#### Pseudocode :-

1. BFS - Early - Stopping ( $G, s, t$ )
2. for each vertex  $v$  in  $V$  do
3.      $Visited[v] \leftarrow FALSE$
4.      $parent[v] \leftarrow NIL$
5.     Create empty queue  $Q$
6.      $Visited[s] \leftarrow TRUE$
7.      $parent[s] \leftarrow NIL, Level[s] \leftarrow 0$
8.      $ENQUEUE(Q, s)$
9. while  $Q$  is not empty do
10.      $u \leftarrow DEQUEUE(Q)$
11.     if  $u == t$  then     // Early Stopping
12.         return Reconstruct\_path( $parent, s, t$ )
13.     for each neighbor  $v$  of  $u$  do
14.         if  $Visited[v] == FALSE$  then
15.              $Visited[v] \leftarrow TRUE$
16.              $parent[v] \leftarrow u$
17.              $level[v] \leftarrow level[u] + 1$
18.              $ENQUEUE(Q, v)$

19. return NOT FOUND // t is unreachable.

Reconstruct\_path (parent, s, t).

1. path  $\leftarrow$  empty list
2. crawl  $\leftarrow$  t
3. while crawl  $\neq$  NIL do
4.     prepend crawl to path
5.     crawl  $\leftarrow$  parent[crawl]
6. return path.

Input :-

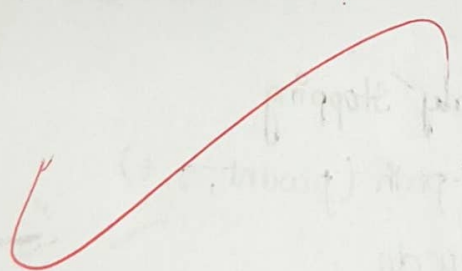
- \* Graph  $G(V, E)$
- \* Start node s
- \* Target node t

Output :-

\* shortest distance from s to t (or path)

If t is not reachable,

return NOT FOUND



# Flowchart (BFS with Early Stopping).

